

A versatile droplet on demand generator based on active pressure control

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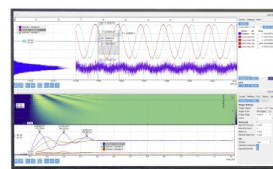
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ABSTRACT

The development of a novel on-demand droplet generator is reported in this paper based on off-the-shelf electromechanical components. With the ability to produce a broad range of droplet sizes, this device was designed to have minimum components without compromising reliability. The generator performance was assessed by producing water droplets for fire dynamics and suppression studies. Droplet formation at the tip of the nozzle was visualized using high-speed imaging. Stable and unstable regions can be identified easily through the images. Interchangeable nozzles can be used, allowing different droplet sizes. The setting parameters of the device allow accurate control of droplet size and initial speed. The device controls the liquid pressure actively and precisely. The active control of pressure and the ability to have specific droplet ejection parameters ensured the repeatability of droplet conditions. Characterization of the device is presented along with the calibration data, which can be collected in an efficient methodology. These data enable the selection of droplet size and speed when required. This system could easily be implemented in other research areas. Hence, the intention is to share and highlight the generator features for its use in a range of other scientific applications.

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I. INTRODUCTION

Droplet generators are essential instruments used in many fields, including printing, spray processes, and surface impact studies.^{1–3} Single droplet experiments are commonly conducted to study droplet formation, evolution, stability, evaporation, surface impact, and combustion.⁴

Drop on demand techniques promise accurate droplet generation and high repeatability. Challenges arise, however, when an attempt is made to utilize current droplet generation devices in various applications. Difficulties such as satellite droplets, shape distortion, bubbles, and trajectory variation are common operation challenges. Therefore, researchers often turn to other experimental approaches where an acceptable compromise is usually adapted.

Repeatable experiments require reliable droplet generation. Existing on-demand droplet generators require careful tuning, while most in house manufactured devices are complex and made for a specific application. Several droplet generator novel designs have

been proposed for a variety of reasons including construction simplification, enhanced control, disturbance reduction, and high extreme working condition ability.

Table I shows recent innovative devices incorporating new ideas and their properties.

Other designs implemented thin piezoelectric elements. Unique techniques were presented in order to reduce the power requirements such as low voltage piezoelectric films for portable devices.¹⁰ The technique was used as a pump at a maximum rate of 3.6 $\mu\text{l}/\text{min}$. Nonetheless, current piezoelectric designs require power amplifiers and stable pulses, and the droplet size has a nonlinear relation with the pulse width. Although the head and piezoelectric casing can be relatively inexpensive,⁹ the required power equipment should also be taken into account. In addition, piezoelectric designs require careful tuning and experimentation to produce the same size droplets, and there are known problems encountered in practice such as voltage wave stability and trapped air.¹¹ They are extremely sensitive to the applied voltage level, frequency, and wave shape. The trapped air can deteriorate the system performance entirely, and

TABLE I. Properties of recently reported droplet generators.

Technique	Droplet size (mm)	Nozzle size (mm) and type	Production rate	Mode	Remarks
Piezoelectric element ⁵	0.65 1.32	1.02 mm 3D printed ABS	Every 2 s	On demand	3D printed structure
Resistive heating of nozzle tip ⁶	0.11 2	0.11 mm–0.31 mm stainless steel syringe tips	Injected by a syringe up to a specific hanging size	Single	Heating the nozzle tip to produce a droplet resulted in shape distortion and secondary droplets occasionally
Mechanical vibration ⁷	0.1 1	0.2 heat processed capillary glass	Every oscillation tested up to 5 kHz	Continuous stream only	Used a hard drive actuator arm to vibrate a capillary tube nozzle
Solenoid impact ⁸	0.1 0.3	0.1–0.3 sapphire	1 Hz to tens of Hz (as stated)	On-demand	Utilized a solenoid as a hammer to provide a pressure pulse
Piezoelectric element ⁹	0.5 1.4	0.5–1.4 stainless steel	Not tested	On-demand	A pump was used to keep a fluid reservoir at the same level
Active pressure control	0.9 1.7	0.5 brass nozzle	5 Hz, tested at 0.9 mm droplet	On-demand and continuous stream	The instrument reported in this paper; production rate depends on droplet size and nozzle

usually, it requires system reassembly and refilling to be in working order again.

In order to provide a solution for combustion experiments and other scientific studies, a novel drop on the demand generator was designed using the shelf components. No component manufacturing was required, which resulted in a precise and straightforward device. The device capability was demonstrated by analyzing the droplet generation process and its utilization in the experimental work.¹² Single droplets were captured using high-speed imaging. The flow parameters such as pressure were controlled by an electronic circuit module along with a microcontroller development board.

II. SYSTEM CONSTRUCTION

Figure 1 shows the components of the instrument, and Fig. 2 illustrates the device electrical and hydraulic circuits. The device consisted of six main components: a liquid reservoir, solenoid valve, nozzle, MPX5100DP pressure sensor, Arduino Uno board, and peristaltic pump. A stepper motor ran the pump, and the pump provided positive displacement flow to the reservoir. The pump type was peristaltic, where a force is applied on a tube to push the flow forward. The components, including the pump, are widely available. The pressure sensor provided a feedback signal to the controller. The signal was acquired by the Arduino development board that is based on the microcontroller ATmega328. The microcontroller sampled the sensor analog signal and provided an output signal to the pump motor, regulating the pressure of the reservoir. Additionally, the Arduino board provided a precise timing control of the solenoid valve. Energizing the solenoid allows the flow to pass through the tubing, and eventually, a droplet was ejected with an initial speed v_i out of a brass nozzle. The nozzles employed in the system were also readily available. Overall, these components enabled the creation of a reliable system that can be assembled, programmed, and operated in a short time.

Reservoir pressure was controlled electronically, and two methods were employed. First, an ON-OFF controller was implemented.

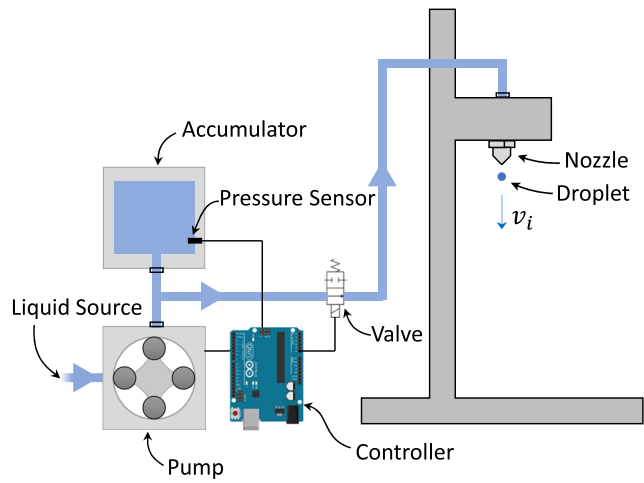


FIG. 1. Components of the device.

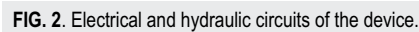


FIG. 3. Block diagram of active pressure control implemented in the controller.

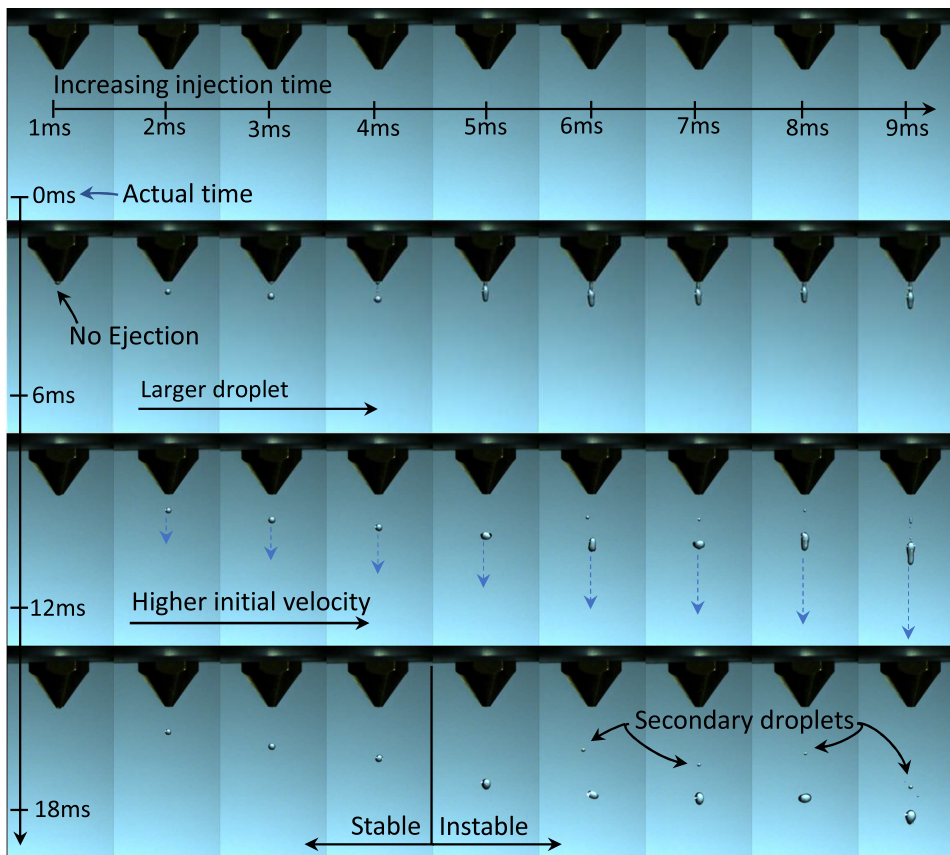


FIG. 4. Evolution of droplets with increasing injection time.

therefore a larger droplet, provided the injection time stays constant. The relation can easily be fit with a polynomial at each pressure and can then be used to determine the required injection time for a specific droplet size. The droplet diameter is not the only

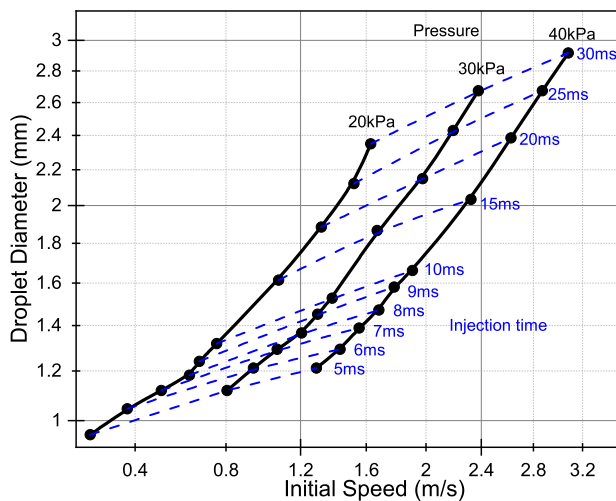


FIG. 5. Droplet size and initial speed increase with injection time at different pressure settings.

parameter that changes with injection time but also the initial speed of droplets.

Figure 5 also presents the relation of the initial speed with injection time. The momentum of the injected flow out of the nozzle is transferred to the generated droplet. The droplet acquires a higher initial speed with increasing injection time. For example, if the nozzle is pointed downward toward a target surface at a distance of 0.2 m from the nozzle, the impact velocity is higher for larger droplets, as shown in Fig. 6. One way to achieve constant impact velocity is to vary the distance between the nozzle and a target surface.

Figure 7 indicates the distance between the nozzle and a surface to attain the same impact velocity with the increasing droplet diameter. Because larger droplets are ejected with higher initial speed, the distance should be reduced between the nozzle and the surface. Therefore, the acceleration due to gravity will be limited by the distance available. Note that this is only true if the droplet does not reach its terminal velocity.

IV. VARIABILITY

In order to study the variability of the droplet generation process, droplet ejection was repeated five times at the same conditions with a pressure setpoint of 20 KPa. Table II lists the injection time tested and the equivalent average droplet diameter. It also shows

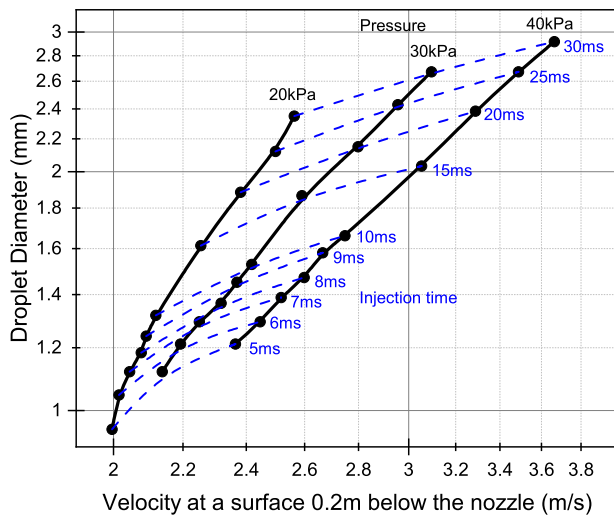


FIG. 6. Droplet velocity at a distance of 0.2 m below the nozzle.

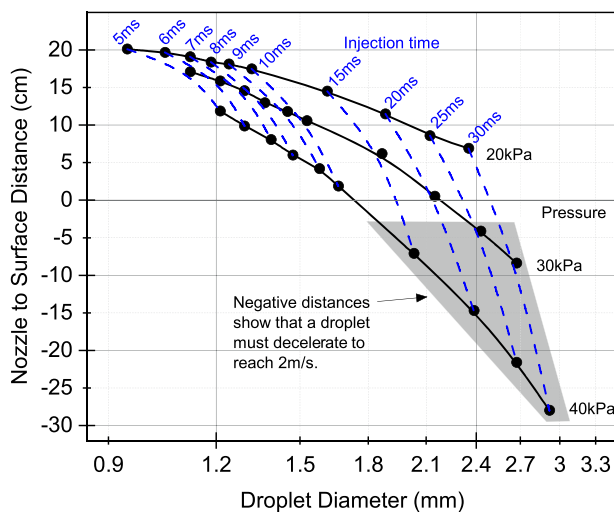


FIG. 7. Required distance from the nozzle to a target to achieve a constant velocity with different droplet sizes.

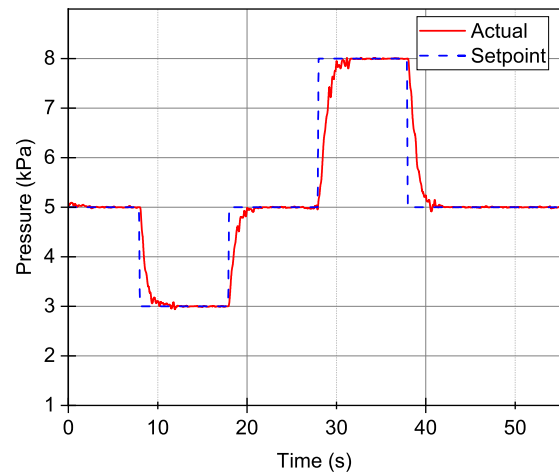


FIG. 8. Setpoint signal and pressure response.

the percentage deviation in droplet diameter and initial speed. The droplet diameter was measured using images scaled to an object of a known size. The initial speed was obtained using a series of droplet images after detachment from the nozzle using the speed $v = d/t$ formula, where d is the distance traveled and t is the time duration between images.

Table II shows that smaller droplets have higher error percentages. Instabilities can affect the measurement as in the case of 1.67 mm diameter where the deviation was high. Error percentages of the initial speed values are considerably lower than error percentages of the diameter values.

V. CONTROL RESPONSE

The results presented in Fig. 8 show the response of pressure control using a PID Controller. The PID performed better than the ON-OFF controller, which suffered from fluctuations at the setpoint. The PID method produced a rapid response when changing the pressure setpoint. These values were dependent on the parameter tuning in the microcontroller code, sensor capability, and setpoint pressure.

It is evident in Fig. 8 that the actual signal takes more time to stabilize when the difference between two consecutive setpoints

TABLE II. Droplet diameter and initial speed deviation at a setpoint pressure of 20 kPa.

Injection time (ms)	Average diameter (mm)	Max. deviation from the average diameter (mm)	Diameter error percentage (%)	Average initial speed (m/s)	Max. deviation from the average initial speed (m/s)	Initial speed error percentage (%)
6	0.93	0.10	10.75	0.742	0.024	3.19
8	1.28	0.06	4.69	0.861	0.019	2.17
10	1.43	0.07	4.90	0.981	0.011	1.15
12	1.54	0.07	4.77	1.060	0.012	1.13
14	1.67	0.16	9.58	1.196	0.007	0.56
15	1.76	0.07	4.17	1.206	0.019	1.60

is more considerable. This delay is a direct result of the response time of the instrument. For example, the controller took 2 s approximately to raise the pressure from 3 kPa to 5 kPa; in contrast, it took 3 s to increase the pressure from 5 kPa to 8 kPa.

A. Pressure setpoint control and tuning

The PID control method was tested for different setpoint profiles implemented in the microcontroller code. This test presented the importance of correct PID tuning. The response could be made faster by removing the differential component of the controller, implementing a PI controller with slight overshoot being tolerated.

Flexibility was an advantage in terms of the pressure control strategy. The control method can be chosen to fit other applications as required. This feature is not limited to the droplet generator mechanism implemented in this system but can be integrated into other droplet generation techniques. Ultimately, the pressure control pump is suitable for other applications where a particular fluid needs to be maintained at specific conditions.

B. Pressure sensor calibration and acquisition

The pressure sensor used was MPX5100DP, a gauge sensor with two inlets. One inlet was by design against fluids as stated in the datasheet. This inlet was connected to the accumulator. The other inlet was left open to the atmosphere. The sensor required no calibration because the additional pressure above atmospheric pressure was measured.

The pressure sensor output was an analog signal; hence, an analog reader or at least an analog to digital converter was required. The microcontroller ATmega328 acquired the signal on the Arduino Uno development board. The advantage of this microcontroller is the ability to read analog signals with 10-bit resolution. In fact, it has five analog inputs, and only one pin was required in the current system. The sensor value was measured every 100 ms and was used as an input to the control algorithm. The same pressure values were sent in real-time to a computer using the serial interface and universal serial bus (USB) port available on the board.

C. Control application

The current droplet generator functions can be significantly enhanced by developing an interface. An application can be developed to control the droplet generation process. The application should allow a fast generation procedure by having a user interface. Parameters such as the pressure setpoint can be directly changed

when needed, and a custom delay time allows a user to set a specific time to produce a particular droplet size. The droplet stream function can also be integrated to give the ability to eject several droplets with a defined delay between consecutive droplets. Pressure monitoring will allow live monitoring of both the setpoint and the actual measured pressure. In addition, the relation between device settings can be mapped against the ejected droplet properties, hence enabling the selection of the required size and initial speed of the ejected droplet.

VI. CONCLUSION

The droplet generator presented in this paper provides an exceptional tool for research studies. It can be used in a wide range of applications requiring pressure regulated stream and repeatable droplet ejection conditions. This generator offers a user-friendly alternative to other systems, which usually require power amplifiers and lengthy tuning. High-speed images were taken of the generated droplets at different conditions. The relation between control parameters and ejected droplets was presented. A control system was implemented to compensate for the lost pressure actively.

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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